

Agentic AI Course Book

Module 1: Foundations of Agentic AI

- What is Agentic AI
- Agents vs Workflows
- Reasoning
- Planning
- Autonomy Levels

Module 2: LLM Fundamentals

- Transformers
- Tokenization
- Embeddings
- Inference
- Context Windows

Module 3: Prompt Engineering

- Role Prompting
- Few-Shot
- Chain of Thought
- ReAct
- Reflection

Module 4: Tool Use

- Function Calling
- APIs
- Search Tools
- Database Tools
- Custom Tools

Module 5: Memory

- Short-Term Memory
- Long-Term Memory
- Vector Memory
- Knowledge Bases

Module 6: Agent Architectures

- Single Agent

- Planner-Executor
- Supervisor
- Hierarchical Agents

Module 7: Agentic RAG

- Retrieval Planning
- Query Rewriting
- Verification
- Grounding

Module 8: Multi-Agent Systems

- Collaboration
- Delegation
- Communication
- Coordination

Module 9: Evaluation & Safety

- Benchmarks
- Guardrails
- Prompt Injection
- Monitoring

Module 10: Production Projects

- Research Agent
- Coding Agent
- Support Agent
- Workflow Agent